
Observe, Attribute, Then Improve: A Critic-Ranked Five-Mechanism Program for an Autoresearch Plateau

GPT Research Coordinator¹ Independent Architecture, Memory, Optimization, and Systems Proposers¹
Independent Adversarial Critic¹

Abstract

This paper prospectively freezes the next five rounds of a 300-second, single-H200 language-model autoresearch campaign after a control-transport failure and a performance plateau. The campaign minimum, 0.927183 validation bits per byte (BPB), remains a quarantined historical treatment observation, not an executable control or a new evidence contribution. Three independent proposal agents read primary 2025–2026 work across normalization, initialization, conditional memory, data order, optimization, and kernels. A separate adversarial critic rescored every candidate on novelty, provenance, causal validity, local impact, transfer reliability, feasibility, falsifiability, and program coherence. The resulting connected program is: (1) exact packer-sourced FA3 boundary metadata, conditional on a robust throughput gate; (2) canonicalized Engram hash keys; (3) conditioned per-head Q/K initialization plus an explicitly local full-rank V adaptation; (4) attention-output Per-RMSNorm, conditional on gradient-safety mediators; and (5) a pack-preserving compression curriculum. The highest experimental score belongs to canonicalized keys; the boundary sidecar is sequenced first because it is exact-semantics infrastructure whose accepted gain would increase charged update exposure for every subsequent round. Each accepted mechanism becomes the comparator for the next, producing one attributed model rather than a bag of additions. A fresh same-seed recovered-recipe comparator

is permitted only within each new causal question. This paper authorizes no endpoint claim by itself: every round must first clear its offline mediator, parity, source, clock, and isolation gates. The governing principle is that before a system can improve itself, it must be able to observe and attribute its failures.

1. Status, Evidence Labels, and Non-Claims

This is the new prospective paper 019. It follows the retrospective GPT_paper_attempt_1, which superseded the former papers 019–022. It may authorize at most five sequential research rounds after typed registry binding, code reconciliation, and green setup checks. It contains no new experimental result.

The public chart starts at experiment 501 and its recorded timestamp, with val_bpb = 0.927183. That scalar was produced by one historical seed-47 treatment under a recovered six-coordinate recipe. It is a provenance and visualization anchor only. A later governed four-pair transport wave was terminally invalid because a foreign process occupied one assigned GPU; the watchdog attributed the exact PID and exited 79. Three completed recovered endpoints—0.927454, 0.930124, and 0.927999—are descriptive only. Thus:

no transport verdict or control rebind,
no new SOTA

The failed wave cannot be retried, remapped, repaired, or subsetted. The new questions below use fresh seeds beginning at 63 and do not estimate recipe transport.

2. Scientific Frame and Comparator Contract

The active challenge is one physical H200 per arm, STOP_MODE=time, TIME_BUDGET=300, FA3, varlen document isolation, a frozen tokenizer/split/evaluator, and max-autotune-no-cudagraphs. The primary estimand is raw, same-seed,

¹OPHIS governed autoresearch campaign.
Correspondence to: GPT Research Coordinator
<baiyuzhu@mit.edu>.

concurrent treatment-minus-comparator endpoint BPB. The effective standard deviation $\sigma = 0.001198$ fixes the minimum useful effect

$$2\sigma = 0.002396.$$

Moving compilation, packing, metadata construction, prefetch, or queue priming outside the charged clock is forbidden unless both arms perform the same work there and the intervention is explicitly defined as changing the producer interface.

Recipe comparator. The within-experiment comparator is current canonical source with WINDOW_PATTERN=TTTL, device/global batch 72/147456, matrix LR .04, warmdown .85, continuous Muon momentum, FA3 varlen/both document isolation, 64× n-gram tables, and every remaining frozen setting. Fresh concurrent execution makes this a valid comparator for

$$Y(\text{recipe} + M) - Y(\text{recipe}),$$

where M is one new mechanism. It does not make the recipe a qualified global control, validate the invalid transport wave, or turn 0.927183 into a contemporaneous observation.

3. Multiagent Generation and Independent Critique

Architecture, memory/optimization/data, and systems/kernel agents generated frozen slates independently. They could inspect the repository and primary sources but could not edit the live experiment graph. A fourth agent then checked source faithfulness, local code fit, arithmetic, causal contrasts, mediator observability, novelty duplication, dependencies, clock placement, and reward-hack surfaces. The critic was told that the ordinary governed scheduler executes exactly two roles per experiment; proposed four-arm designs therefore required decomposition or rejection.

Every vector is scored from 1 (weak) to 5 (strong) in the fixed order: novelty (N), primary-source provenance (P), causal validity (V), expected local impact (I), transfer reliability (R), feasibility/cost (F), numerical falsifiability (X), and connected-program coherence (C). A veto overrides a high sum. Prestige affects P only; it cannot repair weak scope match or arithmetic.

Table 1. Primary provenance of the connected program. Organization and venue raise provenance, never local validity.

source	organization/venue	exact local boundary
FA3	Tri Dao et al.; NeurIPS 2024	kernel semantics only; sidecar efficacy is local
DSE/Engram	DeepSeek-AI, Peking University; ACL 2026 + updated code	canonical keys only; 128k-to-8k transfer unproven
Conditioned init	Australian Institute for Machine Learning, Adelaide; ICLR 2026	Q/K faithful; disjoint V is local
Peri-LN	NAVER Cloud, KAIST, NAVER AI Lab; ICML 2025	missing attention arm; local scale fixed
curriculum	École Polytechnique, MBZUAI; EACL 2026	exact-pack local order only

4. Final Ranking and Why Rank 1 Is Plausible

Canonicalized keys have the highest standalone score. The sidecar is placed first for a dependency reason: current compact_doc_boundaries scans the packed CUDA tensor, clones it, calls data-dependent nonzero, and concatenates the terminal offset inside every charged training step. The packer already knows the same literal boundaries. Transporting immutable, pinned int32 offsets with the batch could remove this synchronization and allocation without changing a single model input, attention segment, target, parameter, or optimizer update. If the measured token-rate gain is large enough, the 300-second frame executes more updates and tokens. For token-law slope b ,

$$\Delta \text{bpb}_{\text{speed}} \approx -b \log r.$$

With measured uncertainty $b \in [0.046, 0.086]$, central break-even is $r = \exp(0.002396/0.06) = 1.04074$, but robust break-even at the low slope is

$$r_{\text{robust}} = 1.05347.$$

Therefore the earlier $r = 1.045$ forecast is insufficient under uncertainty: it predicts only -0.002025 at $b = 0.046$. Rank 1 is plausible only because its semantics can be proven identical and because a strict profile can kill it before a BPB run.

5. Candidate-Specific Offline Gates

5.1. Round 1: packer-sourced FA3 boundary metadata

Before scoring, 1000 literal packed batches must match bitwise on inputs, targets, epochs, cu_seqlens, maximum sequence length, boundary counts, and a boundary digest. Storage must be pinned int32; producer lifetime must extend through a consumer-stream

Table 2. Architecture and systems slates. “Proposer → critic” vectors are N/P/V/I/R/F/X/C. All proposed ideas retain explicit scores.

candidate	proposer → critic	critic sum	verdict and principal reason
Peri attention-output RMSNorm	3/5/5/4/4/5/5/5 3/5/4/3/3/4/5/5	→	32 keep ; strong ICML evidence, but zero-c_proj makes gradient safety a hard gate
Conditioned Q/K + disjoint-V extrapolation	4/4/3/3/2/5/5/5 4/4/3/3/2/5/5/5	→	31 keep ; zero runtime cost; V differs explicitly from the repeated-head paper construction
TTTL local-RoPE/global-NoPE	4/5/3/2/1/5/5/5 3/4/2/1/1/5/5/4	→	25 kill ; long-context scope mismatch and sub-gate short-context prediction
full Differential Attention	5/4/3/3/2/3/5/5 4/5/2/2/1/1/5/3	→	23 kill ; 6–12% optimized-kernel penalty and overlap with current head normalization
Memory+	5/5/3/4/1/2/5/4 4/5/2/3/1/1/5/3	→	24 defer ; bootstrap, runtime, and scale dependencies exceed this block
mHC	5/4/3/4/2/2/5/4 4/4/2/2/1/1/5/3	→	22 kill/defer ; duplicates an unexecuted local direction and source-faithful scope is large
Q/K spectral correction	3/5/3/2/2/5/5/3 3/4/2/1/1/5/5/2	→	23 kill ; dominated by zero-runtime conditioned initialization and conflicts with local QK norm
FA3 packer boundary sidecar	4/4/5/3/4/4/5/5 4/4/5/3/3/4/5/5	→	33 keep ; exact semantics, conditional on robust isolated-H200 throughput
physical packing of 14 n-gram tables	4/4/5/4/3/4/5/5 4/3/4/2/2/3/5/4	→	27 defer ; profile-dependent child, not a first mechanism
radix/segmented sparse RMSProp	5/4/4/5/2/1/5/5 4/3/2/2/1/1/5/3	→	21 defer ; kernel dependency and adjacent archived idea
shape-aware HiMuon	4/3/4/4/3/3/5/4 4/3/2/2/1/3/4/3	→	22 kill ; throughput-quality arithmetic is below the BPB gate
stack-backed Muon layout	4/3/5/2/4/2/5/5 3/2/4/1/3/2/5/4	→	24 defer ; profile bundle only after a measured bottleneck
persistent end-to-end FP8	3/5/3/5/2/2/5/4 1/5/2/1/1/2/5/2	→	19 kill ; dependency and numerical-risk veto

Table 3. Memory, optimizer, and data-order slate. Vectors are N/P/V/I/R/F/X/C.

candidate	proposer → critic	critic sum	verdict and principal reason
small batch + token clocks	3/5/5/4/5/4/5/5 3/3/2/3/2/2/4/4	→	23 kill ; source supports Adam β_2 , not the bundled Muon/RMSProp extrapolation; four-arm scheduler mismatch
mass-conserving memory router	5/5/4/4/4/4/5/5 3/3/3/2/2/3/4/4	→	24 kill ; synthesis, adjacent gate history, and 1.5% cost consume the low forecast
pack-preserving compression curriculum	3/5/4/4/5/4/5/5 3/5/4/3/3/3/5/5	→	31 keep ; source-grounded ordering mechanism with exact-pack isolation required
per-row hit-clock RMSProp	5/4/4/4/3/3/5/5 4/2/3/2/1/1/3/4	→	20 defer ; sparse-kernel and bias-correction dependencies
canonicalized Engram keys	4/5/4/3/5/5/5/5 4/5/4/3/3/5/5/5	→	34 keep ; highest experimental score; local key-collapse mediator is mandatory
curriculum × LR	4/5/4/4/4/3/5/5 3/4/3/3/2/4/3/4	→	26 defer ; conditional child only after the ordering mediator is supported
blockwise sharpness diagnostic	3/5/3/4/4/3/5/4 3/4/3/1/2/3/3/5	→	24 offline ; useful failure attribution, but not a model-improvement round
Memory Grafting	5/4/2/5/2/1/4/4 5/4/2/3/1/1/1/4	→	21 defer ; donor construction must be charged end to end; current form has a clock/reward-hack veto
MeCo metadata conditioning	3/5/2/2/4/3/4/3 3/5/1/1/2/4/3/2	→	21 kill ; source validation PPL loses to its longer baseline and the target mismatches local BPB
CLIMB mixture optimization	4/5/2/4/3/1/4/3 3/5/1/2/2/1/1/2	→	17 kill/defer ; data already use climbmix; iterative selection is heavy and exposes validation leakage

Table 4. Connected execution order after critic review. The raw critic sum is not the only dependency criterion.

round	mechanism	score / 40
1	FA3 boundary sidecar	33
2	canonicalized Engram keys	34
3	conditioned Q/K + local V	31
4	attention Peri-RMSNorm	32
5	compression curriculum	31

event. Variable boundary counts must produce exactly one compiled graph after warmup. A 20-step equal-step comparison must leave model parameters and optimizer state bitwise equal. On a fresh isolated H200, the point token/s ratio must be at least 1.054 and its 95% lower bound at least 1.041. Failure kills the round

without an endpoint score.

5.2. Round 2: canonicalized Engram keys

DeepSeek’s Engram demo maps token strings through NFKC, NFD, accent stripping, lowercasing, and whitespace normalization before n-gram hashing (Cheng et al., 2026; DeepSeek-AI, 2026). The proposed arm changes memory keys only; model input IDs, targets, bytes, document boundaries, and token order remain fixed. Special, BOS, and replacement-character IDs remain one-to-one. At least 10% of local key cardinality and 10% of observed n-gram occurrences must be affected; at least 95% of alias sets must be explained by the declared transformations. Non-memory initialization and global RNG must match

bitwise, charged slowdown must be at most 0.5%, and compiled graph count must be unchanged. The intrinsic prediction is -0.0030 BPB; a 0.5% cost gives an approximate net -0.002699 . In the pilot, affected-token byte-weighted BPB must improve by at least .003 while unaffected-token BPB regresses by no more than .0015.

5.3. Round 3: conditioned attention initialization

The ICLR 2026 source uses independent semi-orthogonal per-head Q/K maps and the same rectangular-identity V construction for every head (Saratchandran and Lucey, 2026). The local fused full-rank identity instead gives disjoint V coordinate blocks across heads. It is therefore named and scored as an extrapolation, not a paper-faithful V replication. Required preflight conditions are: Q/K row-orthogonality residual at most 10^{-5} ; V $\sigma_{\min}/\sigma_{\max} \geq 0.99$; every non-Q/K/V parameter and global RNG bitwise equal; initialized attention-Jacobian surrogate down at least 20% in every layer; attention entropy change at most .05; finite, nonzero step-2 Q/K/V gradients; and median step time within .5%. The raw paired prediction is at most -0.0030 BPB. Failure of the mediator kills the arm before scoring.

5.4. Round 4: attention-output Peri-RMSNorm

Peri-LN reports consistent larger-scale loss improvements and attributes them to normalizing both sides of residual sublayers (Kim et al., 2025). OPHIS already implements the MLP-side form; this round adds fixed, scale-free RMS normalization only after attention `c_proj`. The local `c_proj` is zero initialized. Step-0 value parity is thus uninformative: after the first update, RMS normalization could amplify a tiny branch. We require step-0 residual output exactly zero, step-1 `c_proj` gradient-norm ratio in $[0.5, 2.0]$, no layer above $2.5\times$ control, and no non-finite value. Across a 32-step diagnostic, residual-norm coefficient of variation and max/min layer-gradient ratio must both fall at least 20%. Slowdown must be at most 1%. The distinction between fixed OPHIS RMS normalization and the source’s generally learned scale is part of the claim record, not hidden in implementation.

5.5. Round 5: pack-preserving compression curriculum

The EACL 2026 study trained more than 200 models and reports 18–45% earlier or mid-training step savings, with compression ratio among the strongest difficulty signals (Zhang et al., 2026). The local arm isolates order: one immutable prepadded cache must give

identical whole-epoch pack-ID, token, target, crop, BOS, byte, and boundary multisets. Compression score must separate top and bottom deciles by at least 0.5 pooled SD. A complete ten-bin interleaving schedule is frozen and hashed before launch; adaptive checkpoint selection is forbidden. Loader/step-time difference must be at most .5%. The final prediction is raw $\Delta\text{bpb} \leq -0.0040$; the 25%-budget probe must improve by at least .002 and the final endpoint must still clear the main gate. An early-only win rejects transfer to the actual objective.

6. Sequential Funnel, Combination Rule, and GPU Policy

For each eligible round, stage 1 is one new pair. Promotion requires raw $\Delta\text{bpb} \leq -0.002396$, every integrity invariant, and the candidate-specific mediator. It remains a screen, not an effect claim. At $n = 3$, require mean ≤ -0.002396 , 3/3 helping, and a one-sided 95% upper bound below zero. At $n = 6$, require mean ≤ -0.002396 , at least 5/6 helping, and upper bound ≤ -0.001198 . At $n = 10$, require mean ≤ -0.002396 , at least 9/10 helping, and upper bound ≤ -0.002396 , followed by one never-used test evaluation. Candidates are never ranked by their stage-1 means.

An accepted mechanism becomes the comparator for the next round. A rejected mechanism leaves the cumulative comparator unchanged. Thus:

$$R_0 \rightarrow R_0 + M_1 \rightarrow R_0 + M_1 + M_2 \rightarrow \dots,$$

with every arrow prospectively attributed. Interaction tests require a later paper; no accidental combination is interpreted.

GPU occupancy is subordinate to causal integrity. Stage 1 uses one pair and keeps the protocol’s one-to-two-GPU reserve. A promoted $n = 3$ stage can use three concurrent pairs across six tenant-free GPUs. No unrelated four-question/all-eight wave is allowed merely to maximize utilization: doing so would recreate Q1’s attribution exposure and violate the sequential combination rule. Two clean UUID/PID inventories, immutable assignments, continuous co-tenancy sampling, and fail-closed per-arm watchdogs are mandatory.

7. Reproducibility and Failure Attribution

For every selected source, the campaign must preserve the official paper or landing-page identity, retrieval metadata, SHA-256, atomic claims, claim scope, limitations, evidence assessment, and official code snapshot when available. DeepSeek’s Engram demo is code

provenance for canonicalization, not proof of a local BPB effect. Conditioned initialization and Peri-LN have archived official proceedings PDFs. Curriculum uses the official ACL Anthology artifact. The sidecar is a local systems mechanism and must bind its equivalence harness and profiler rather than borrow efficacy from FlashAttention-3 (Dao et al., 2024).

Every code change is default-inert, fail-closed, echoed in `RESOLVED_CONFIG`, bound to the intervention record, tested locally, synced to the governed remote runtime, SHA-matched, and reconciled before a new challenge-selection event. Each endpoint closes through RunRecords, EvidenceRecords, EvidenceUpdates, and Decisions. Failure reports must name which link broke: source, config, initialization, mediator, compile graph, clock, device/PID isolation, completion, or statistical gate.

8. Explicit Casualties and Limits

The critic hard-vetoes every Q1 retry or “qualified 0.927 control” claim. RoPE/NoPE, Differential Attention, mHC, FP8, HiMuon, and the current mass-router formulation fail scope, runtime, arithmetic, duplication, or dependency checks. The small-batch source directly supports Adam second-moment token half-life, including $0.95^{147456/262144} = 0.9715597802$, but does not support simultaneously transporting that rule to Muon and sparse RMSProp (Marek et al., 2025). Its proposed four-arm form is also not executable by the ordinary scheduler. Physical table packing, sparse kernels, stack layouts, per-row clocks, curriculum $\times$ LR, and Memory+ remain deferred children. Blockwise sharpness remains an offline attribution diagnostic.

Ratings are judgments, not outcomes. Canonicalized keys may collapse too few of an 8192-token vocabulary. Sidecar bookkeeping may be too small to clear robust break-even. Conditioned initialization may be neutral under zero-c_proj; Peri normalization may destabilize that same bootstrap. Curriculum gains may vanish at the final rather than early endpoint. These are not caveats to explain away afterward; they are the pre-registered falsifiers.

9. Conclusion

The plateau response is broad but not indiscriminate. Independent agents generated ideas from strong 2025–2026 primary work; an adversarial agent penalized extrapolation, duplication, weak local scope, and impossible scheduling. The result is a five-round dependency body with explicit scores, mechanisms, mediators, kill gates, and a valid concurrent recipe com-

parator. The system will first prove that it can observe what changed and attribute why a round failed. Only then may a lower BPB become evidence of improvement.

References

- T. Dao et al. FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision. NeurIPS, 2024.
- X. Cheng et al. Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models. ACL, 2026. doi:10.18653/v1/2026.acl-long.226.
- DeepSeek-AI. Engram Architecture Demo Implementation. https://github.com/deepseek-ai/Engram/blob/fb7f84a21f91223715394a33a1dc24bbfb7f788e/engram_demo_v1.py, 2026.
- J. Kim et al. Peri-LN: Revisiting Normalization Layer in the Transformer Architecture. ICML, 2025.
- M. Marek, S. Lotfi, A. Somasundaram, A. G. Wilson, and M. Goldblum. Small Batch Size Training for Language Models: When Vanilla SGD Works, and Why Gradient Accumulation Is Wasteful. NeurIPS, 2025.
- H. Saratchandran and S. Lucey. Conditioned Initialization for Attention. ICLR, 2026.
- Y. Zhang, A. Mohamed, H. Abdine, G. Shang, and M. Vazirgiannis. Beyond Random Sampling: Efficient Language Model Pretraining via Curriculum Learning. EACL, 2026.